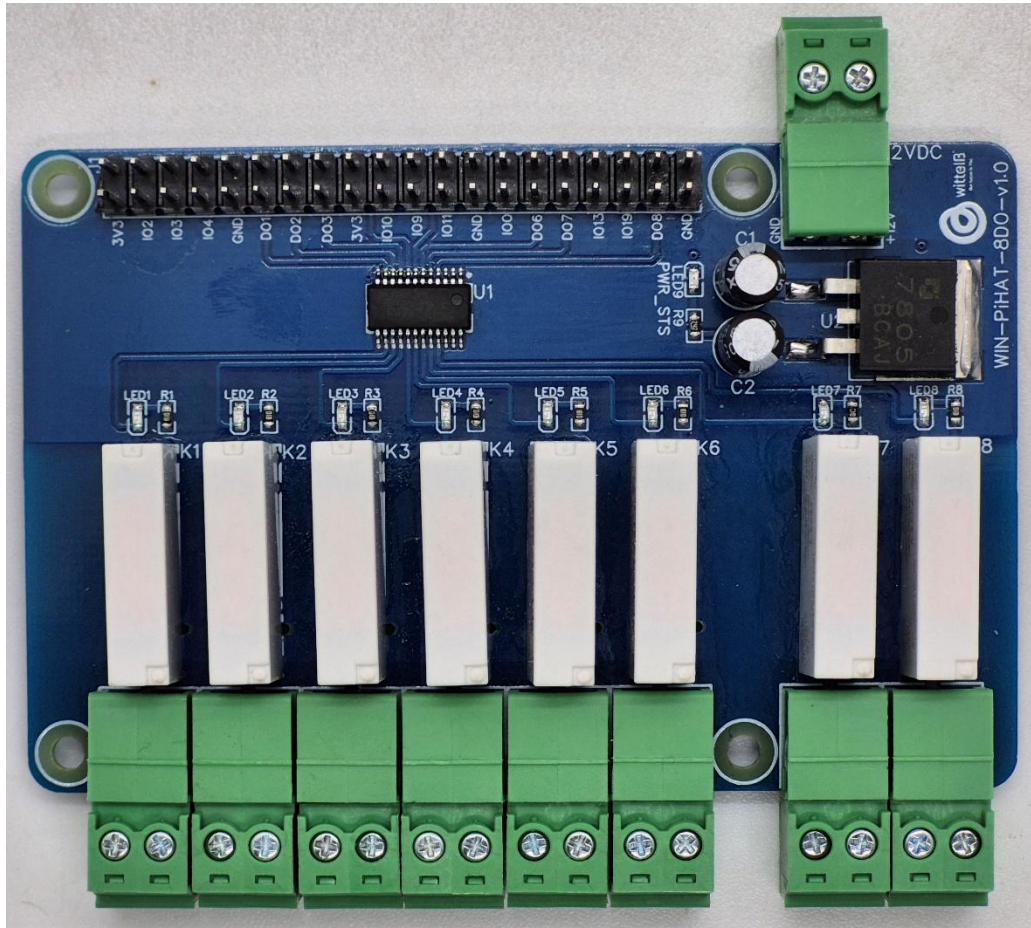


Technical Datasheet of WIN-PiHAT-8DO

1. Overview

The **Digital Output (DO) PiHAT** is an 8-channel expansion board for Raspberry Pi 4 and CM4, offering **relay-isolated digital outputs** for **industrial load control**. It protects your system by providing **electrical isolation** between external loads and the Pi's GPIO header, ensuring reliable operation in industrial environments.



PiHAT features a **scalable architecture** supporting **1 to 8 independent relay-driven digital outputs (DO)**. Its **population-based design** allows channel flexibility while maintaining a **compact and reliable solution** for embedded control, automation, and industrial load switching applications.

Specifications

Category	8DO PiHAT
Output Type	Relay-Isolated Digital Outputs, 5V SPST-NO Industrial Relay, 5A @ 250VAC / 30VDC
Channel Count	8Channels (Expandable Design Architecture)
Connectors	9 of 2.54 mm Pluggable Terminal Blocks (2-Pin), 40-Pin Raspberry Pi GPIO Header

General :-

Dimensions	56 mm L x 85 mm W
Power	Input Power – 12/24V DC
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DO (Digital Outputs) with Relay :-

Channels	8
Contact Form	1A (SPST-NO)
Contact Material	AgNi Alloy
Contact Capacity	5A @ 250VAC, 5A @ 30VDC
Coil Voltage	5VDC
Coil Power	0.12W
Insulation Resistance	≥100MΩ
Electrical Life	1 × 10 ⁵ operations
Mechanical Life	1 × 10 ⁷ operations
Operating Time	≤10 ms (max., at rated voltage)
Release Time	≤05 ms (max., at rated voltage)

Configuration :-

All 8 DO channels operate as relay-driven digital outputs and are interfaced to the Raspberry Pi GPIO header through a Darlington transistor array. Output control is performed via software running on the Raspberry Pi. Each channel is configured as a GPIO output and drives a corresponding relay.

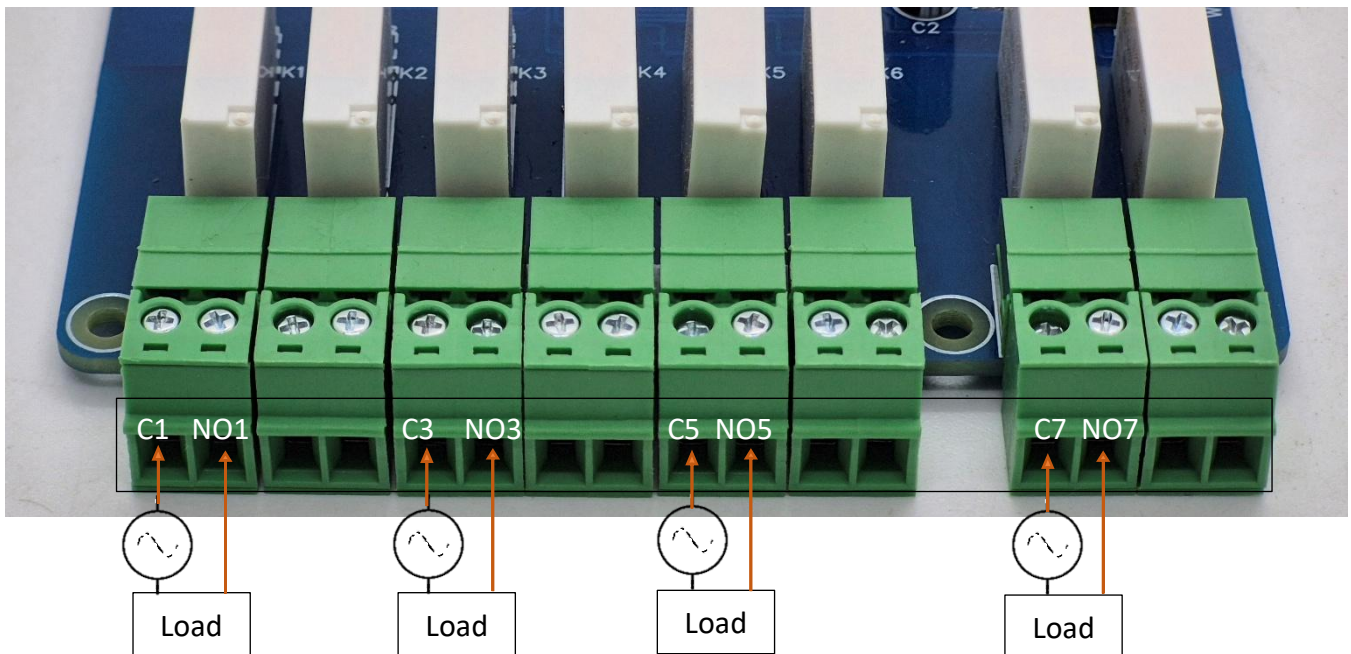
When a GPIO pin is driven HIGH, the Bipolar Transistor sinks current through the relay coil, energizing the relay and closing the Normally Open (NO) contact. When the GPIO pin is driven LOW, the relay coil is de-energized and the contact returns to its default open state.

The following are step by step guides for configuration:

1. Connect the Module to +12/24V DC POWER SUPPLY.
2. Connect the Board to the Raspberry Pi via header pins

Sr. No.	Digital Input	Pi GPIO	Header Pin number
1	D01	17	11
2	D02	27	13
3	D03	22	15
4	D04	24	18
5	D05	25	22
6	D06	05	29
7	D07	06	31
8	D08	26	37

Wiring Diagram:-

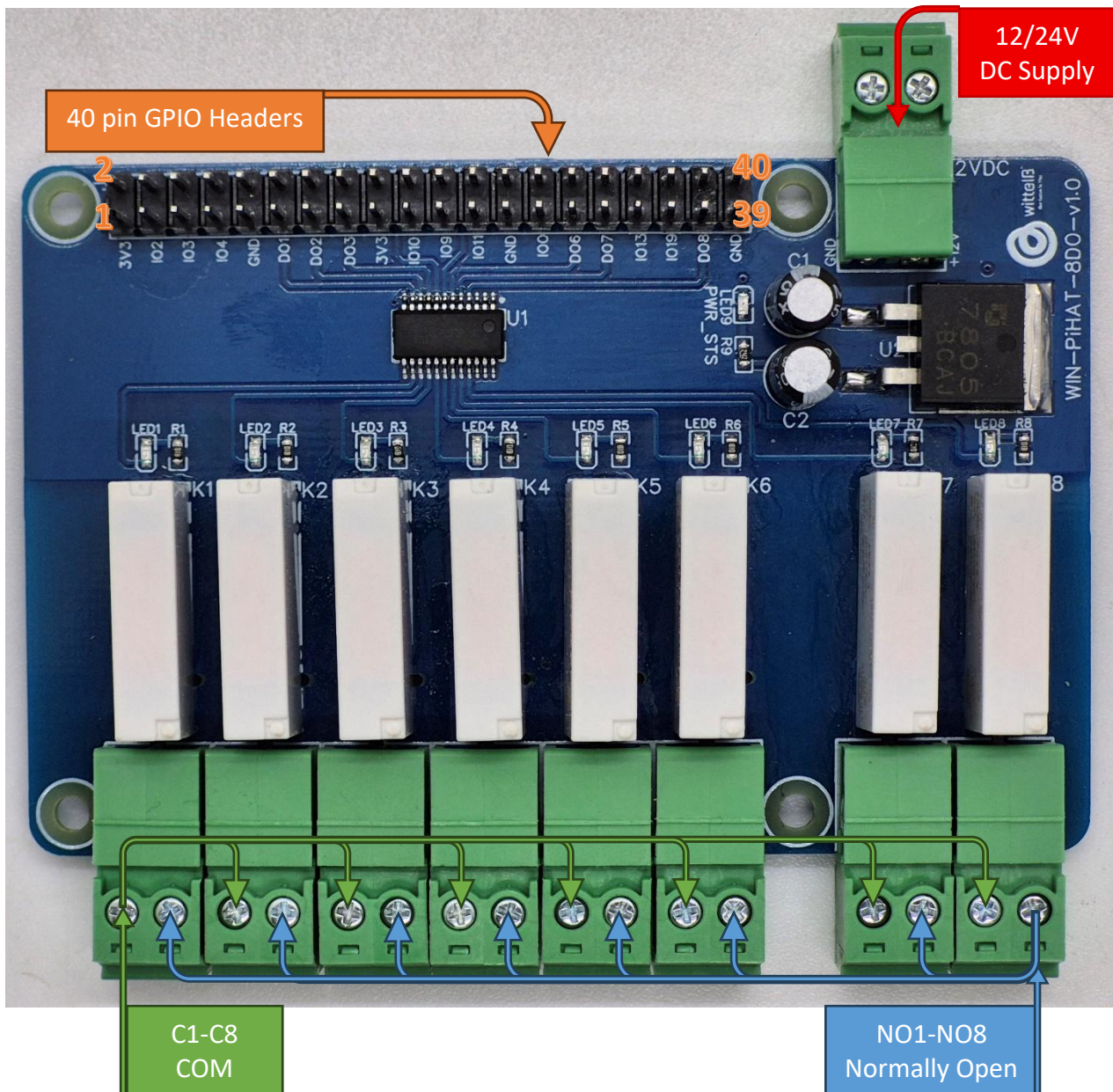


- For Digital Output (DO) operation, each channel provides two terminals: C (Common) and NO (Normally Open). When the relay is energized, terminal C is internally connected to NO. When the relay is de-energized, the connection between C and NO is open.
- For AC load connection, the AC Live (Phase) wire is connected to terminal C, and the load is connected between terminal NO and Neutral. For DC load connection, the positive supply (+V) is connected to terminal C, and the load is connected between terminal NO and Ground.
- It is recommended to switch the Live (Phase) conductor in AC applications. Ensure that the connected load does not exceed the relay rating of 5A 250VAC or 5A @30VDC.

Sample Test Code:

- [WIN-PiHAT-8DO\(Relay\)](#)

Hardware Interface & Connector Layout:



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